

Unit Conversions

Everyday microfluidics units — a Semitree cheat sheet

Quick conversions for the quantities you juggle most in the lab. All values follow directly from SI definitions.

Flow rate

$$1 \mu\text{L}/\text{min} = 16.67 \text{ nL}/\text{s} = 0.06 \text{ mL}/\text{h}$$

$$1 \text{ mL}/\text{h} = 16.67 \mu\text{L}/\text{min} ; 1 \mu\text{L}/\text{min} = 1.667 \times 10^{-11} \text{ m}^3/\text{s}$$

Pressure

$$1 \text{ bar} = 100 \text{ kPa} = 1000 \text{ mbar} = 14.504 \text{ psi}$$

$$1 \text{ psi} = 6.895 \text{ kPa} ; 1 \text{ mbar} = 100 \text{ Pa}$$

Volume

$$1 \text{ mL} = 1000 \mu\text{L} = 10^6 \text{ nL}$$

$$1 \mu\text{L} = 1000 \text{ nL} = 10^{-9} \text{ m}^3$$

Viscosity

$$1 \text{ mPa} \cdot \text{s} = 1 \text{ cP} = 10^{-3} \text{ Pa} \cdot \text{s}$$

$$\text{water} \approx 1 \text{ mPa} \cdot \text{s} \text{ at } 20^\circ\text{C}$$

Length

$$1 \text{ mm} = 1000 \mu\text{m} = 10^6 \text{ nm}$$

$$1 \mu\text{m} = 10^{-6} \text{ m}$$

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Standard, published equations. Always verify against primary sources for critical work.